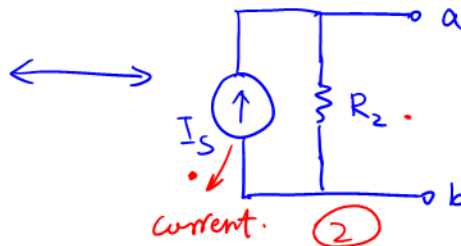
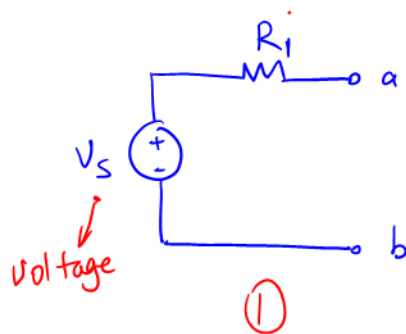
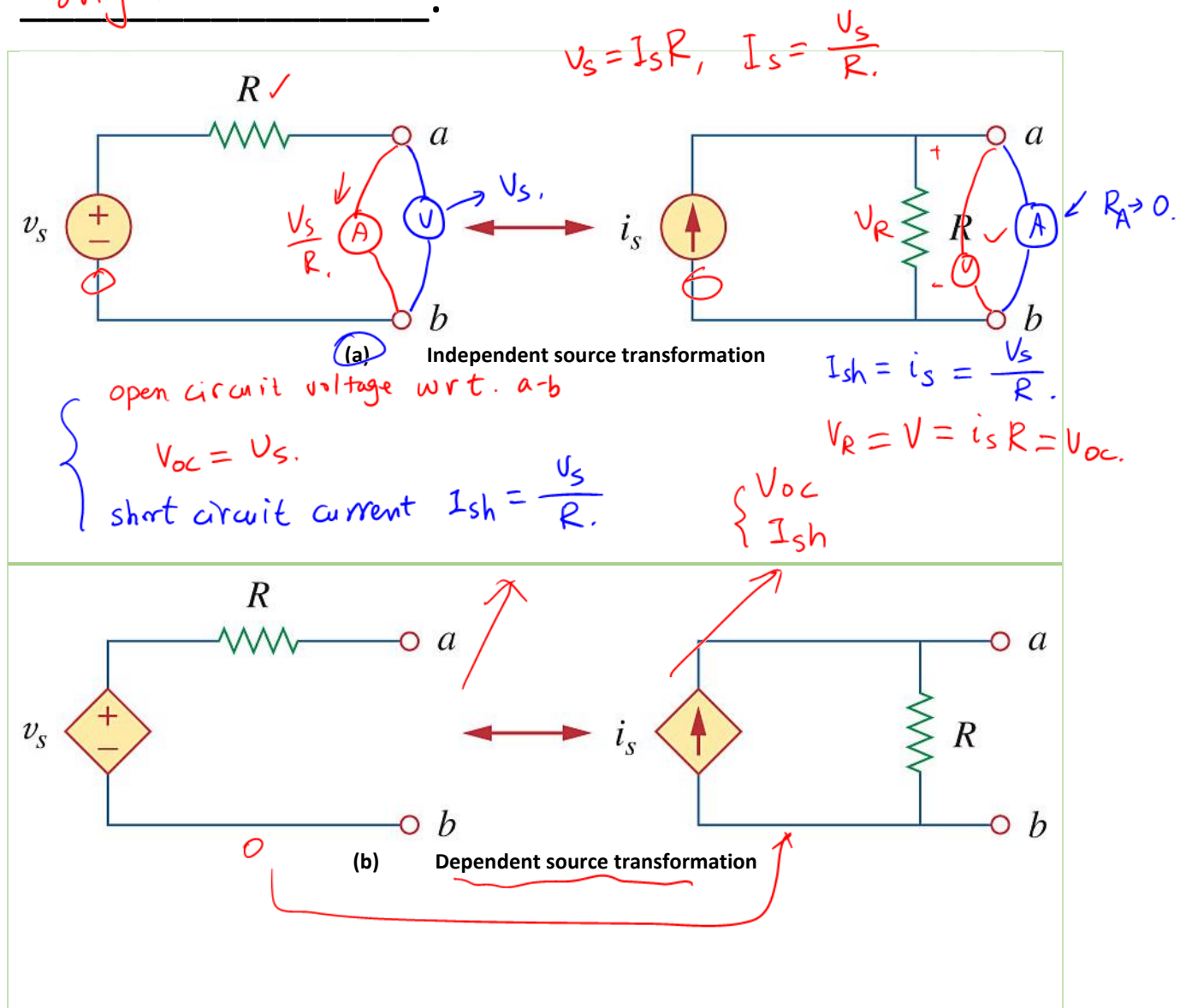


L#17: Source transformation

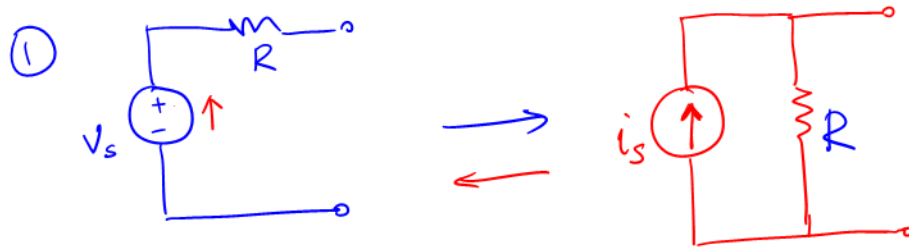


$$\begin{aligned} R_1 &= R_2, \\ I_s &= \frac{V_s}{R_1}, \\ V_s &= I_s R_2. \end{aligned}$$

Equivalent circuit: a new circuit whose electric characteristics are identical with the original circuit.



Methodology:



- get correct circuit configuration.



- get polarity and direction of current correct.

- Arrow of current always points the positive polarity of the voltage source.

Applicability:

- get values of sources and resistance correct.

- value of resistance remains.

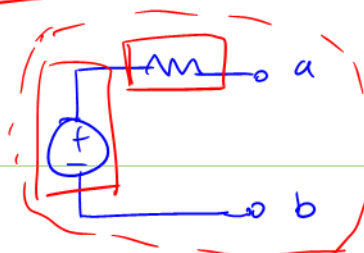
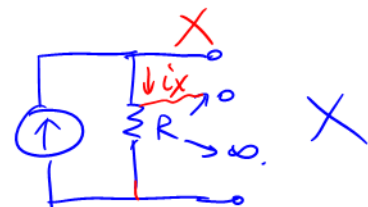
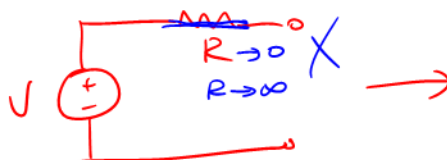
- $i_s = \frac{V_s}{R}$, $V_s = i_s R$, make sure after the transformation, the two circuits are equivalent!

- it's applicable for both dep. and ind. sources.

- $(V) \rightleftharpoons (A)$

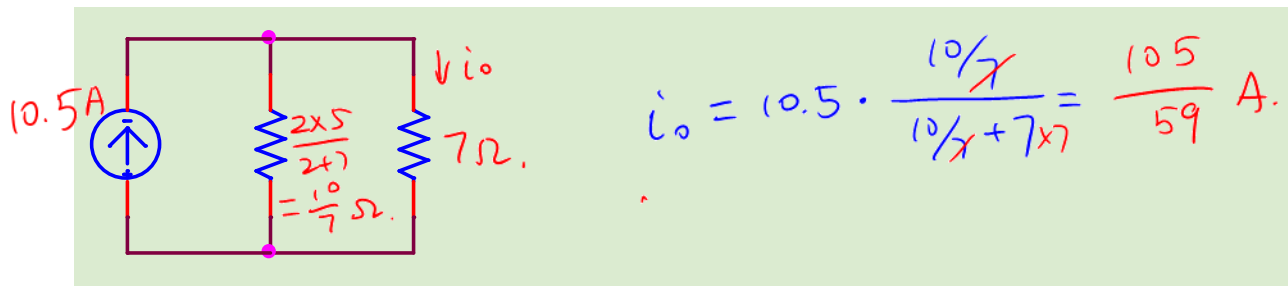
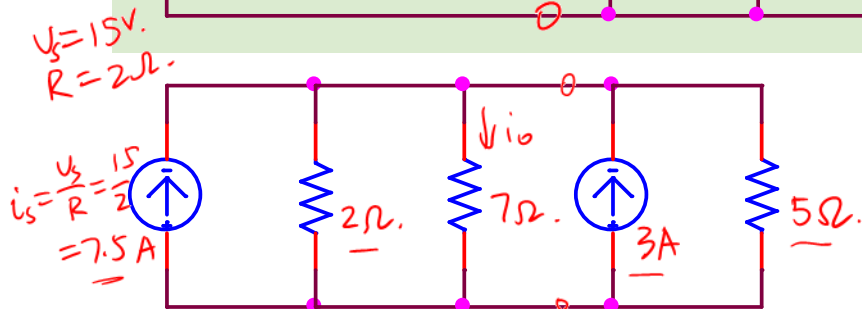
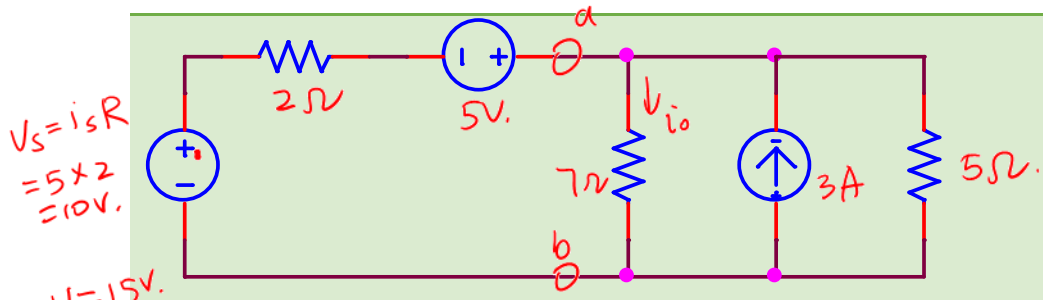
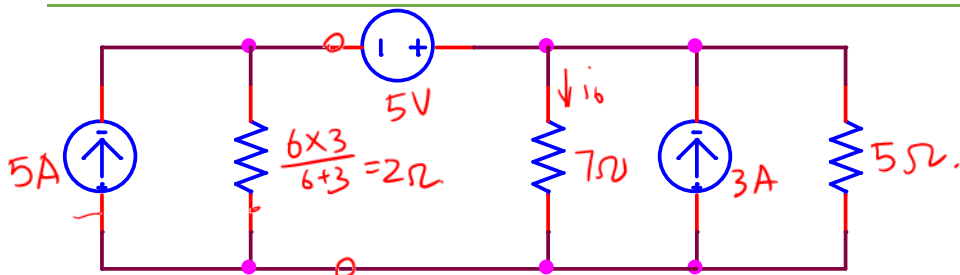
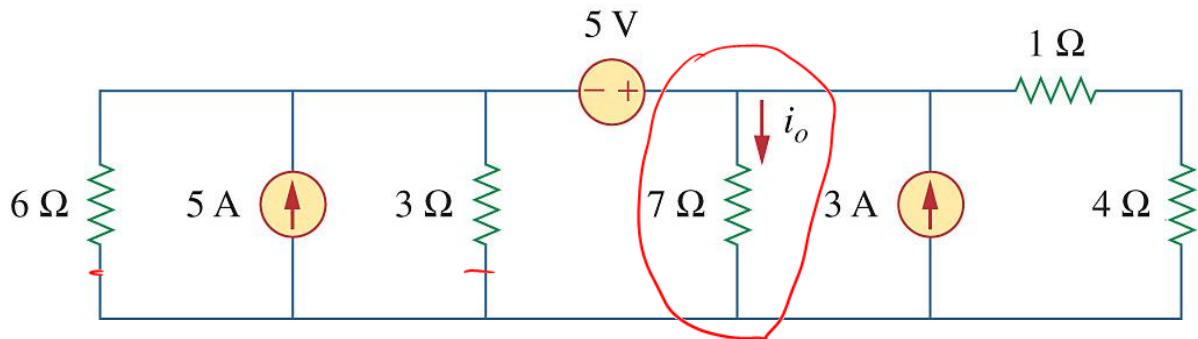
- the boundary that we conduct source transformation.

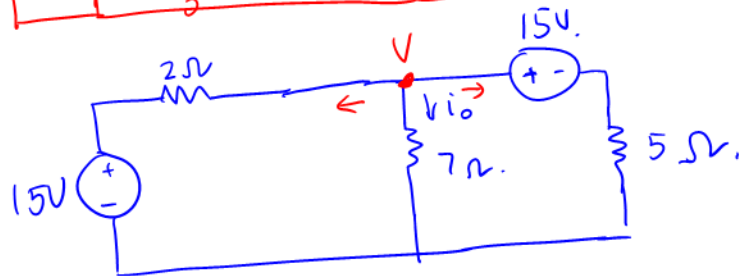
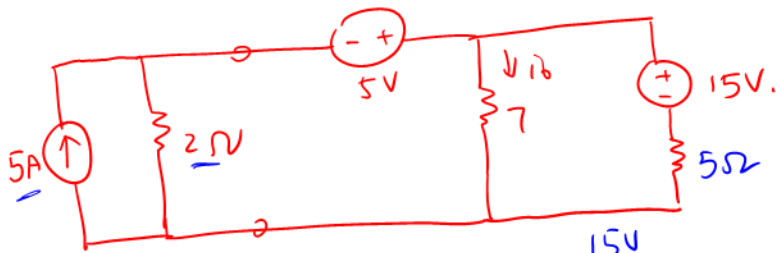
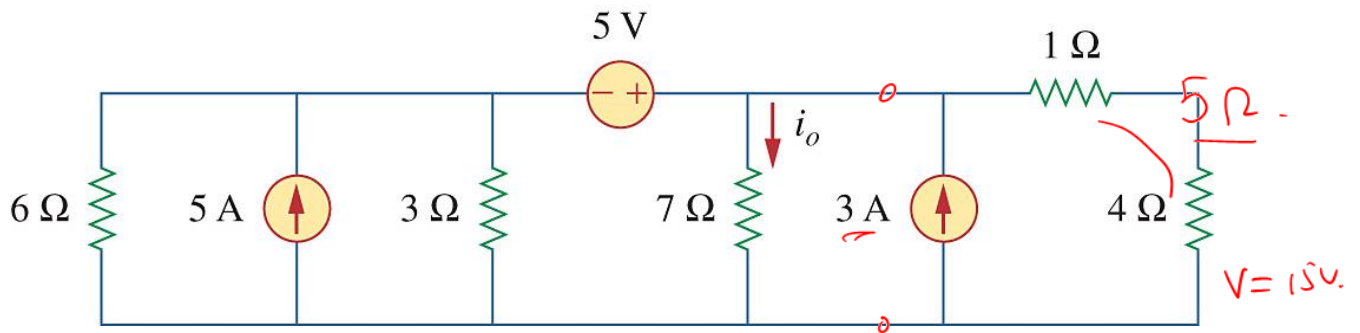
- $\updownarrow$ make sure the correct circuit configuration



- never transform the circuit areas with unknown parameters.

Example 5: Use source transformation and determine i_o .





$$\frac{V-15}{2} + \frac{V}{7} + \frac{V-15}{5} = 0$$

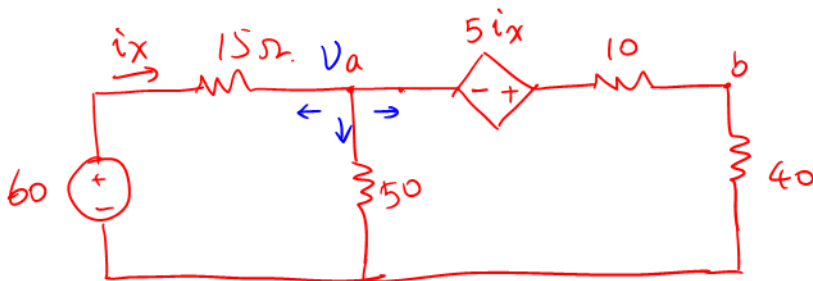
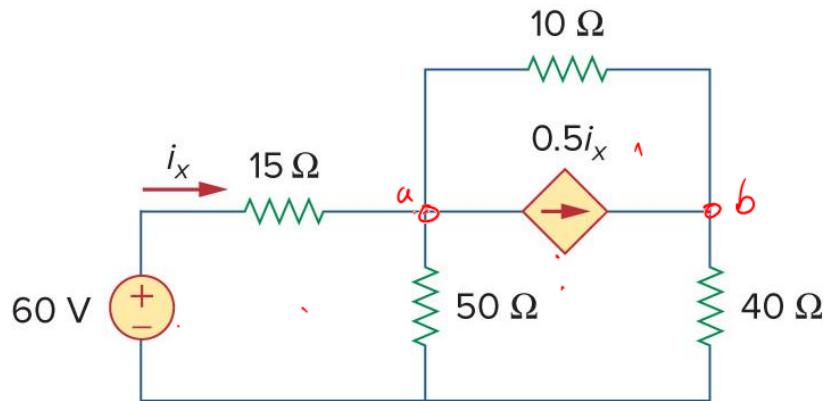
$$\left(\frac{V}{2} + \frac{V}{7} + \frac{V}{5}\right) = 10.5$$

$$(35+10+14)V = 10.5 \times 70$$

$$V = \frac{105 \times 7}{59} \text{ V.}$$

$$i_o = \frac{V}{7} = \frac{105}{59} \text{ A.}$$

Example 6: Use source transformation to solve for i_x .



$$\frac{V_a - 60}{15} + \frac{V_a}{50} + \frac{V_a + 5 \cdot \frac{60 - V_a}{15}}{50} = 0 \quad (1)$$

$$(1) \times 150: 10V_a - 600 + 3V_a + 3V_a + 60 - V_a = 0.$$

$$15V_a = 540 \Rightarrow V_a = 36V.$$

$$i_x = \frac{60 - 36}{15} = 1.6A.$$

Do NOT transform the part of the circuit with the unknown variable!

Example 7: Use source transformation to solve for v_o .

